

Concept Proposal: Non-Invasive Human–Machine Interface for Assistive Exoskeletons

Mateja Glišović

Faculty of Technical Sciences
University of Novi Sad



Bojan Bićanin

School of Electrical Engineering
University of Belgrade



February 2026

Independent Student Research
Collaboration

Core Engineering Challenge

The Translation Gap Between Biology and Machinery

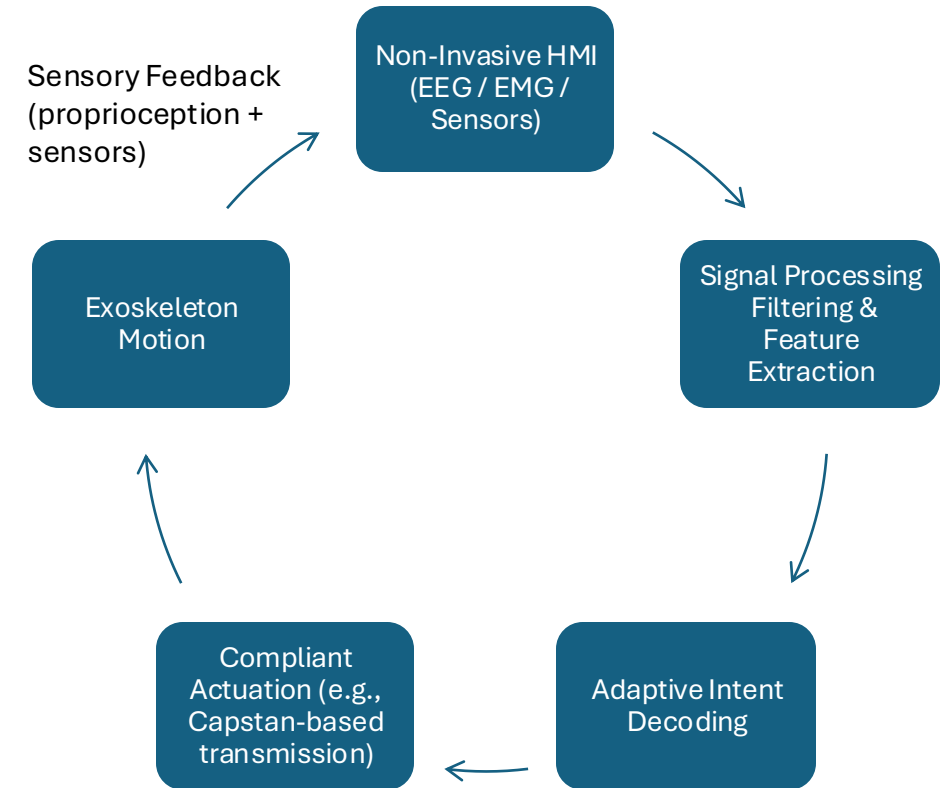
- Non-invasive neural & muscle signals are noisy, weak, and highly variable
- Human movement is adaptive and non-linear, not deterministic
- The exoskeleton can exert forces beyond safe human tolerance
- Latency above ~100 ms disrupts the natural sensorimotor loop
- Safety constraints dominate all control decisions

Engineering Questions

- How can user intention be decoded reliably from non-invasive signals in real time?
- How can perceived latency be reduced below the threshold of natural disruption?
- How can stability and safety be maintained under uncertain and time-varying inputs?

Proposed Direction:

- Combining "Real-time Tracking" and "Predictive Motion Motifs".
- Balancing Mechanical Transparency with functional precision.



Research Context & Inspiration

1. SAN Lab (Tsukuba) – The Foundation

- *Key Insight:* **Cybernic Voluntary Control (CVC)** using bio-signals.
- *Relevance:* Established the importance of the bio-feedback loop.
- *My Takeaway:* The goal is "Fusion," not just "Control."

2. Robinson Lab (Rice) & Motif Neurotech – The Methodology

- *Key Insight:* Neural Dynamics & "Motifs". Treating brain states as network patterns rather than just raw signals.
- *Relevance:* Suggests that modeling neural dynamics as “motifs” may enable predictive control strategies (patterns).
- *My Takeaway:* Applying "Motif theory" to non-invasive motor control could be the key to solving latency.

3. My Proposed Contribution (The Synthesis)

- *The Goal:* To experimentally explore predictive intent decoding and compliant actuation in a controlled subsystem.
- *Current Strategy:*
 - Develop a simulation model of the control loop (HMI → decoding → actuation → feedback) with mechanical kinematics.
 - Analyze latency sensitivity and stability constraints
 - *Build a small-scale physical prototype (hand module) utilizing Dyneema to test 'Cybernic' compliance in hardware.*